

MOSSEL BAY, South Africa

WMO: 689280

Lat: 34.190S

Lon: 22.132E

Elev: 59

StdP: 100.62

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	8.8	-0.1	3.8	18.3	1.8	4.3	16.7	13.1	15.2	10.3	15.6	3.5	290	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.8	27.3	18.2	25.7	19.1	24.6	19.2	22.0	24.4	21.3	23.7	20.7	23.1	3.4	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	15.9	23.4	20.5	15.2	22.9	19.8	14.6	22.4	64.8	24.7	62.3	23.8	60.3	23.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.0	9.3	7.7		6.7	33.1	0.5	2.5	6.3	34.9	6.0	36.4	5.7	37.8	5.4	39.6
				4.6	23.1	0.5	0.9	4.2	23.8	3.9	24.3	3.6	24.7	3.2	25.4
				DB											
				WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	20.9	20.8	19.7	17.8	16.7	14.9	14.5	15.2	16.6	17.8	19.8	(6)	
(7)		DBStd	3.10	1.54	1.59	1.74	2.47	2.33	2.38	2.75	2.21	2.28	1.89	1.69	1.81	(7)
(8)		HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)
(9)		HDD18.3	651	1	1	6	36	62	107	124	121	98	61	29	7	(9)
(10)		CDD10.0	2712	337	302	302	236	208	148	141	141	157	204	234	303	(10)
(11)		CDD18.3	321	79	69	49	21	12	4	6	3	5	6	13	52	(11)
(12)		CDH23.3	636	107	97	67	88	64	26	32	22	38	20	11	63	(12)
(13)		CDH26.7	101	3	8	7	27	17	4	4	5	15	6	2	3	(13)
(14)	Wind	WSAvg	3.4	3.6	3.4	3.2	3.0	2.8	3.4	3.3	3.4	3.6	3.7	3.6	(14)	
(15)	Precipitation	PrecAvg	379	26	23	28	41	34	31	29	39	41	33	25	(15)	
(16)		PrecMax	718	170	84	70	136	130	103	89	168	139	121	157	72	(16)
(17)		PrecMin	184	0	1	2	3	2	3	2	7	0	4	2	0	(17)
(18)		PrecStd	99	31	16	19	32	29	19	21	38	25	31	28	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.7	27.5	27.2	30.0	28.8	27.0	26.8	26.6	28.1	27.1	24.4	26.6	(19)
(20)		0.4%	MCWB	21.7	22.0	19.8	18.3	16.4	14.2	14.3	14.3	15.8	18.1	18.8	21.0	(20)
(21)		2%	DB	25.5	25.4	24.9	25.5	24.5	23.3	23.6	21.4	22.2	22.2	22.7	24.9	(21)
(22)		2%	MCWB	21.2	20.9	19.8	17.0	15.3	13.0	13.2	13.4	15.0	17.2	18.1	20.6	(22)
(23)		5%	DB	24.5	24.5	23.6	22.8	21.5	20.8	20.8	19.2	19.6	20.6	21.6	23.9	(23)
(24)		5%	MCWB	20.6	20.4	19.6	17.1	14.8	12.6	12.7	13.4	14.9	16.7	17.5	19.9	(24)
(25)		10%	DB	23.6	23.5	22.6	21.1	19.8	18.9	18.6	17.6	18.3	19.6	20.8	23.0	(25)
(26)		10%	MCWB	20.1	20.0	19.1	17.1	14.9	12.8	12.6	13.4	14.6	16.2	17.1	19.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.8	23.1	21.8	20.1	18.0	16.3	16.0	16.6	18.0	19.3	20.3	22.3	(27)
(28)		0.4%	MCDWB	25.2	26.2	24.7	24.9	22.2	19.6	19.7	19.4	22.5	22.0	23.2	25.0	(28)
(29)		2%	WB	22.0	21.9	20.9	19.1	17.0	15.3	15.0	15.7	16.7	18.2	19.2	21.2	(29)
(30)		2%	MCDWB	24.3	24.2	23.4	22.2	20.5	18.5	18.8	17.8	19.3	20.7	21.6	23.9	(30)
(31)		5%	WB	21.3	21.2	20.3	18.3	16.5	14.5	14.3	14.9	15.9	17.5	18.4	20.5	(31)
(32)		5%	MCDWB	23.5	23.5	22.7	21.0	19.5	17.8	17.8	17.5	18.6	19.9	20.6	23.1	(32)
(33)		10%	WB	20.8	20.6	19.7	17.7	16.0	13.8	13.7	14.2	15.1	16.8	17.7	19.9	(33)
(34)		10%	MCDWB	22.9	22.8	21.9	20.1	18.5	17.1	17.1	16.7	17.8	19.0	20.1	22.4	(34)
(35)	Mean Daily Temperature Range	MDBR	4.8	4.9	5.0	5.9	6.4	7.0	6.9	6.6	6.1	5.3	5.2	5.1	(35)	
(36)		5% DB	MCDBR	5.7	6.3	6.6	9.4	9.9	10.2	10.4	9.5	8.3	6.9	6.3	6.3	(36)
(37)		5% DB	MCWBR	3.1	3.4	3.4	4.1	4.1	4.4	4.4	4.3	4.2	3.6	3.6	3.4	(37)
(38)		5% WB	MCDWR	4.8	5.4	5.7	6.9	7.0	7.4	7.5	7.0	6.9	5.5	5.6	5.5	(38)
(39)		5% WB	MCWBR	2.8	3.1	3.2	3.7	3.8	4.1	4.3	3.9	3.9	3.3	3.5	3.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.371	0.374	0.360	0.336	0.322	0.310	0.312	0.339	0.351	0.372	0.360	0.368	(40)	
(41)		taud	2.500	2.496	2.524	2.574	2.591	2.596	2.584	2.502	2.450	2.425	2.495	2.495	(41)	
(42)		Ebn at Noon	966	942	918	886	846	831	848	868	912	932	970	973	(42)	
(43)		Edh at Noon	114	110	100	85	74	69	73	89	106	117	114	115	(43)	
(44)	All-Sky Solar Radiation	RadAvg	7.48	6.68	5.46	4.06	3.03	2.60	2.85	3.67	4.99	6.11	7.20	7.75	(44)	
(45)		RadStd	0.27	0.22	0.26	0.18	0.16	0.15	0.16	0.15	0.24	0.33	0.31	0.42	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page